

CAPSTONE PROJECT

PHASE-3

AutoViz AI

“Intelligent Data Analysis and Dynamic Dashboard Generator”

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ABSTRACT

In today's data-driven digital era, organizations generate vast volumes of structured and semi-structured data in multiple formats such as Excel spreadsheets, CSV files, Word documents, and JSON datasets. However, extracting meaningful insights from heterogeneous data sources requires technical expertise, manual preprocessing, and specialized analytical tools. This often creates delays in decision-making and limits data accessibility for non-technical users.

This project, titled “**AutoViz AI – Intelligent Data Analysis and Dynamic Dashboard Generator,**” presents an AI-powered web-based analytics system designed to automatically transform raw multi-format data into interactive and meaningful visual insights. The system enables users to upload datasets in various formats, after which automated data extraction, preprocessing, and exploratory data analysis (EDA) are performed.

AutoViz AI intelligently detects data types, handles missing values, removes duplicates, performs statistical analysis, and recommends appropriate visualization techniques based on dataset characteristics. Interactive dashboards are dynamically generated using advanced visualization tools, allowing users to explore insights through bar charts, pie charts, histograms, scatter plots, box plots, line graphs, and correlation heatmaps.

The system is developed using Python as the core programming language, with FastAPI for backend processing and Streamlit for frontend interface development. Pandas and NumPy are used for data cleaning and transformation, while Plotly enables interactive, theme-aware visualizations. The application also supports light and dark mode themes, along with PDF report export functionality for offline analysis and documentation.

The modular and scalable architecture of AutoViz AI allows for future enhancements such as predictive analytics integration, real-time data streaming, cloud deployment, and role-based dashboard customization. By combining artificial intelligence, automated preprocessing, and interactive visualization within a unified platform, the system reduces manual effort, enhances analytical accuracy, and promotes data-driven decision-making across business, education, and research domains.

Keywords:

Artificial Intelligence, Automated Data Analysis, Data Visualization, Interactive Dashboard, Exploratory Data Analysis (EDA), Multi-Format Data Processing, Business Intelligence, Web-Based Analytics, FastAPI, Streamlit, Plotly, Data Preprocessing, Decision Support System.

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